

Practical 3: Classification using Naive Bayes Algorithm in WEKA

Objective: Demonstrate the classification rule process on WEKA dataset using Naive Bayes algorithm and analyze the generated results.

Dataset Used:

```
@relation weather
```

```
@attribute outlook {sunny,overcast,rainy}  
@attribute temperature {hot,mild,cool}  
@attribute humidity {high,normal}  
@attribute windy {TRUE,FALSE}  
@attribute play {yes,no}
```

```
@data
```

```
sunny,hot,high,FALSE,no  
sunny,hot,high,TRUE,no  
overcast,hot,high,FALSE,yes  
rainy,mild,high,FALSE,yes  
rainy,cool,normal,FALSE,yes  
rainy,cool,normal,TRUE,no  
overcast,cool,normal,TRUE,yes  
sunny,mild,high,FALSE,no  
sunny,cool,normal,FALSE,yes  
rainy,mild,normal,FALSE,yes
```

Step 1: Open WEKA Software

1. Click on Start Menu.
2. Search for WEKA.
3. Open **WEKA GUI Chooser**.
4. Click on **Explorer** option.
5. WEKA Explorer window will open.

Step 2: Create Dataset File

1. Open Notepad.
2. Copy the provided ARFF dataset.
3. Click File → Save As.
4. Save file name as **weather.arff**.
5. Select Save as type → All Files.
6. Click Save.

Step 3: Load Dataset in WEKA

1. Open the **Preprocess** tab.
2. Click **Open File** button.
3. Browse the location of weather.arff.

4. Select the file.
5. Click Open.
6. Wait until dataset loads successfully.

Step 4: Verify Dataset Information

1. Observe relation name displayed in WEKA.
2. Verify total number of attributes.
3. Verify number of instances.
4. Click each attribute individually.
5. Observe histogram and value distribution.
6. Verify class attribute is correctly loaded as **play**.

Step 5: Perform Data Preprocessing

1. Observe all attributes for missing values.
2. If preprocessing is needed, use filters.
3. Click **Choose** under Filter section.
4. Select filters such as:
 - ReplaceMissingValues
 - Normalize
 - Remove
5. Click Apply if any filter is used.
6. Ensure dataset is clean before classification.

Step 6: Open Classification Tab

1. Click on **Classify** tab.
2. This section is used for classification algorithms.
3. Different classifiers are available in WEKA.

Step 7: Select Naive Bayes Algorithm

1. Click on **Choose** button.
2. Navigate to:
bayes → NaiveBayes
3. Select NaiveBayes classifier.
4. The algorithm name will appear beside Choose button.

Step 8: Configure Naive Bayes Parameters

1. Click on the algorithm name **NaiveBayes**.
2. Configuration window will open.
3. Observe available parameters.
4. Enable or disable kernel estimation if needed.
5. Keep default settings for basic execution.
6. Click OK.

Step 9: Select Evaluation Method

1. Under **Test Options**, select evaluation method.
2. Available options include:
 - Use training set
 - Supplied test set
 - Cross-validation
 - Percentage split
3. Select **10-fold Cross Validation**.
4. This method divides dataset into 10 parts for evaluation.

Step 10: Configure Output Options

1. Enable output prediction if needed.
2. Select plain text output format.
3. Observe prediction output settings.
4. Verify correct class attribute selection.

Step 11: Execute Naive Bayes Classifier

1. Click the **Start** button.
2. WEKA starts training the classifier.
3. Probability calculations are performed internally.
4. Dataset instances are classified.
5. Wait until execution completes.

Step 12: Observe Classification Output

1. Output appears in the result panel.
2. Observe correctly classified instances.
3. Observe incorrectly classified instances.
4. Check accuracy percentage.
5. Observe Kappa statistics.
6. Analyze Mean Absolute Error and Root Mean Squared Error.

Step 13: Analyze Confusion Matrix

1. Scroll to the confusion matrix section.
2. Observe actual versus predicted classes.
3. Analyze misclassified records.
4. Compare YES and NO predictions.
5. Evaluate overall model performance.

Step 14: Analyze Precision and Recall

1. Observe detailed accuracy by class.
2. Analyze Precision values.
3. Analyze Recall values.
4. Observe F-Measure.
5. Analyze ROC Area values for classifier performance.

Step 15: Visualization of Results

1. Right-click the result in Result List.
2. Select **Visualize classifier errors**.
3. Scatter plots and prediction graphs will appear.
4. Observe correctly and incorrectly classified instances visually.

Step 16: Final Observation

1. Dataset was loaded successfully.
2. Naive Bayes classifier executed successfully.
3. Classification accuracy was generated.
4. Confusion matrix displayed successfully.
5. Class prediction analysis completed successfully.

Step 17: Conclusion

Classification using Naive Bayes algorithm was successfully implemented in WEKA. The classifier predicted the target class based on probability calculations and generated performance evaluation metrics successfully.